Propeller-Assisted Robust 3D Hopping Robot with Hierarchical Force Allocation

Chuhan Zhang[1,2,3], Hongbo Zhang[1], Yanlin Chen[1,4], Yunxi Tang[1,4], Yun-Hui Liu[1], Mingyi Liu[2,3], and Xiangyu Chu[1,4]
Presenter: Chuhan Zhang - Johns Hopkins University, M.S.E. Robotics czhang.daniel@gmail.com ▪ chuhandanielzhang.github.io
[1]The Chinese University of Hong Kong ▪ [2]Guangdong Technion–Israel Institute of Technology ▪ [3]Technion–Israel Institute of Technology ▪ [4]Multiscale Medical Robotics Centre

SCAN: DEMO VIDEO PAPER + ARXIV

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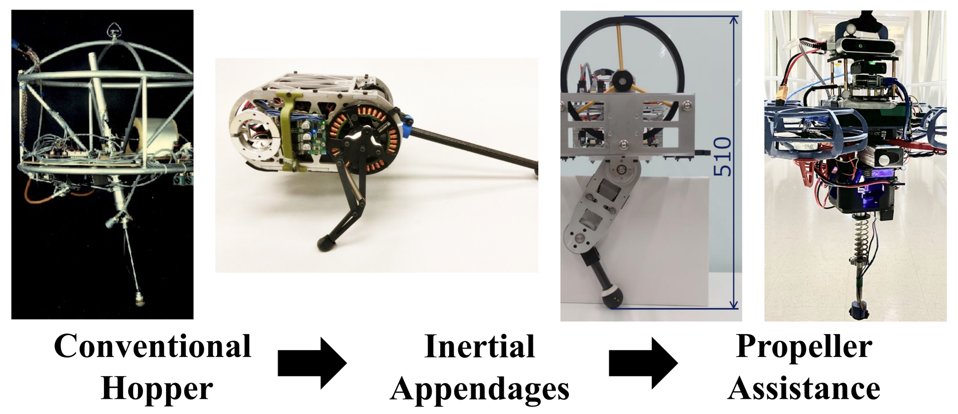
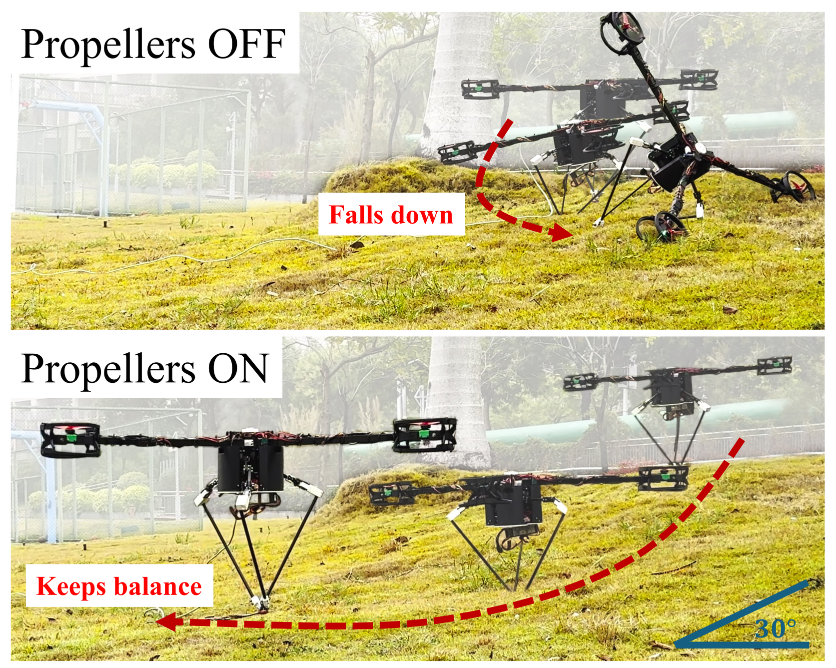
1 Problem - hybrid dynamics create an authority gap



- **Stance**
Ground reaction force exists only during a short contact interval.
- **Flight**
No contact: attitude becomes underactuated.
- **Disturbance**
Impacts and pushes inject attitude and horizontal momentum.

Control authority disappears exactly when attitude can keep drifting.
Design question: how should leg contact force and rotor thrust be coordinated across stance and flight?

2 First proof - stable on 30° grass

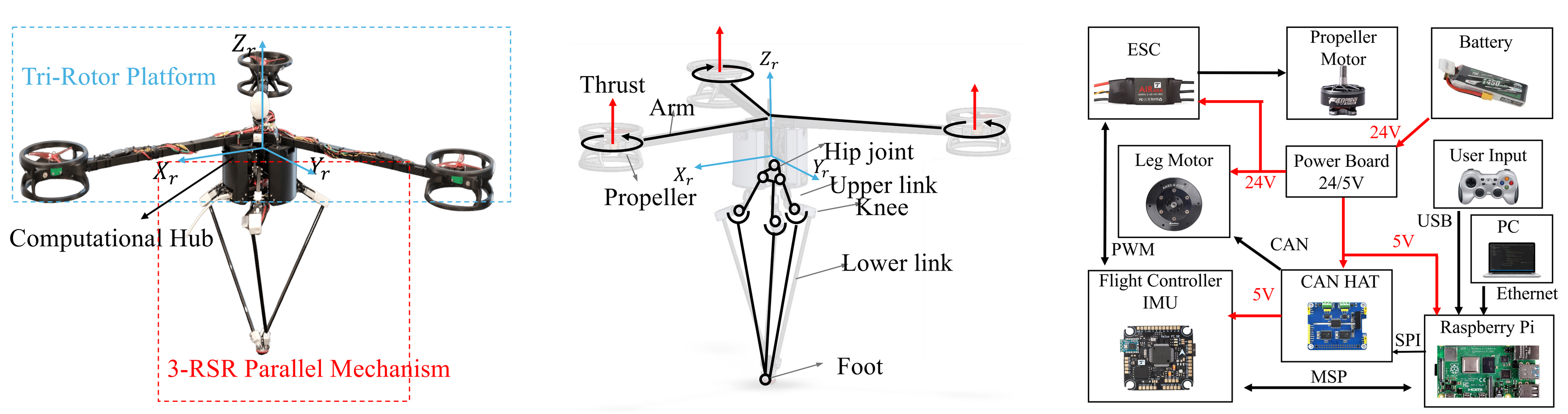


- **Propellers off**
After asymmetric contact, attitude error grows. With no flight-phase actuator, the robot falls.
- **Propellers on**
Differential thrust remains available through flight and restores attitude before the next landing.
- **What this establishes**
Residual authority in stance.
Primary attitude authority in flight.
Faster recovery from uneven impacts.

Not primary lift: the rotors close the attitude-authority gap.

Propellers off: attitude diverges and the robot falls. **Propellers on:** it remains upright and keeps hopping.

3 Platform - hardware built for leg-first hopping

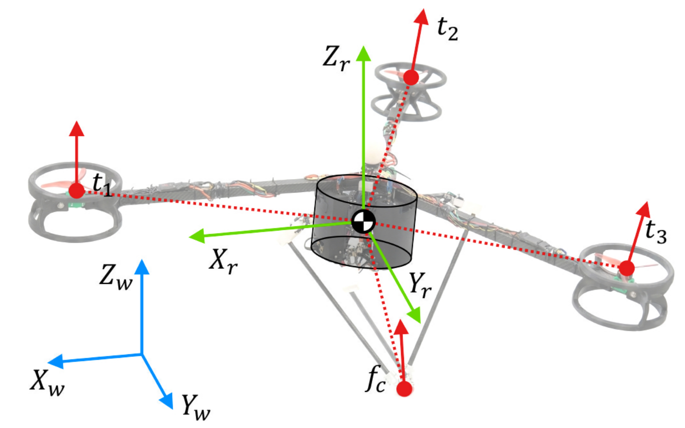


Robot design   Leg and tri-rotor components   Electronic architecture

- **3-RSR parallel leg**
Three RSR chains meet at one foot: fast 3-DoF omnidirectional placement, only **0.35 kg** moving mass.
- **External PC @ 500 Hz**
State machine, estimation, wrench planning, and the HFA allocator.
- **Computational hub**
Batteries and electronics near the hip: low center of mass, low inertia.
- **Raspberry Pi interface**
CAN to the three leg motors; MSP to the flight controller.
- **Y-layout tri-rotor**
Continuous attitude authority; **10 N** per rotor, total $0.82\,mg$: a hopper, not a drone.
- **Real-time link**
LCM over Ethernet: RTT ≈ 0.2 ms, 99% < 0.5 ms, within the 2 ms period.

3.75 kg total ▪ 9.2% leg mass ▪ 20 Nm / leg motor ▪ 500 Hz control

4 Model - one rigid body, two actuator sets



Contact force $\boldsymbol{f}_c$ and thrusts $t_{1,2,3}$ produce the body wrench.

- **Net wrench = leg contact + three rotors**

$$\begin{bmatrix}\boldsymbol{F}_{net}\\ \boldsymbol{\tau}_{net}\end{bmatrix} = \begin{bmatrix}\boldsymbol{f}_c\\ \boldsymbol{r}_c\times\boldsymbol{f}_c\end{bmatrix} + \sum_{i=1}^{3}\begin{bmatrix}\boldsymbol{d}_t\\ \boldsymbol{r}_i\times\boldsymbol{d}_t\end{bmatrix} t_i$$

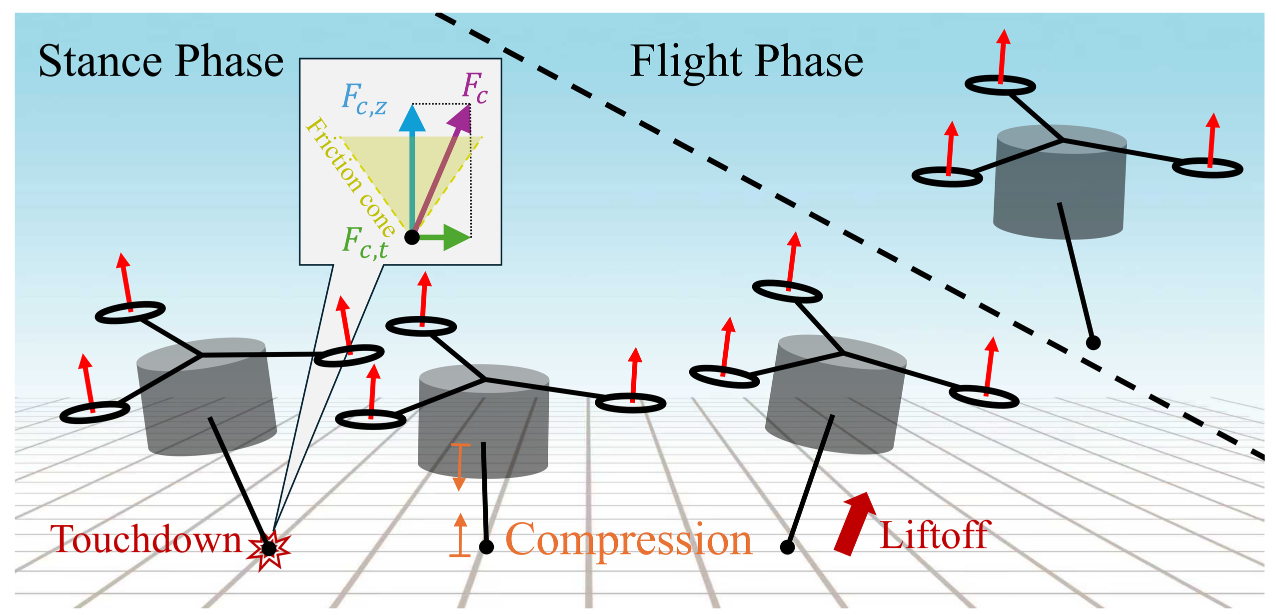
- **Newton-Euler dynamics of the CoM**

$$m\boldsymbol{a} = \boldsymbol{F}_{net} + m\boldsymbol{g}, \quad \dot{\boldsymbol{\omega}} = \boldsymbol{I}^{-1}\left(\boldsymbol{R}^\top\boldsymbol{\tau}_{net} - \boldsymbol{\omega}\times\boldsymbol{I}\boldsymbol{\omega}\right)$$

- **Symbols (values used in all experiments)**

$\boldsymbol{f}_c$ ground reaction force (world)
$\boldsymbol{r}_c$, $\boldsymbol{r}_i$ arms: CoM → contact, rotor i
$t_i \in [0, 10]$ N thrust along $\boldsymbol{d}_t = \boldsymbol{R}\boldsymbol{e}_3$
$\boldsymbol{F}_{net}$, $\boldsymbol{\tau}_{net}$ net force, moment
m=3.75 kg; $\boldsymbol{g}$ gravity; $\boldsymbol{I}$ inertia
$\boldsymbol{R} \in \mathrm{SO}(3)$ attitude; $\boldsymbol{\omega}$ body rate
$\boldsymbol{J}$ leg Jacobian; τ_{max}=20 Nm limit
μ=0.4 friction coefficient
p_z, $\dot{p}_z$ CoM height, velocity; h_{des}=0.664 m
l leg length; l_0=0.464 m; Δ_{td} threshold
k_z, b_z, k_E vertical, k_v, k_r Raibert gains
$\boldsymbol{K}_R$, $\boldsymbol{K}_\omega$ attitude PD gains

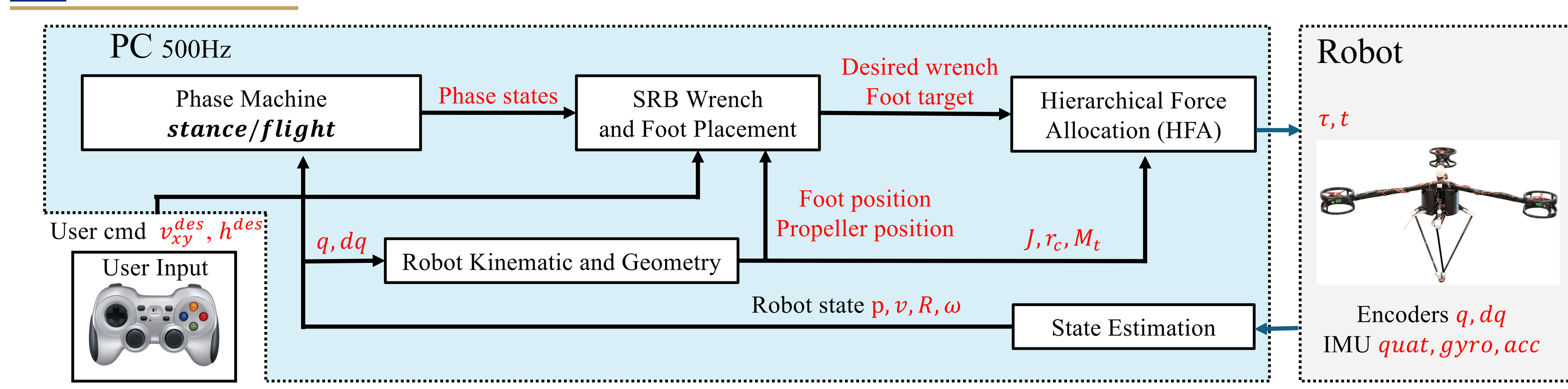
5 One hop - event-driven authority handoff



- **Touchdown:** $l \le l_0 - \Delta_{td}$
Leg absorbs impact; energy law drives the next apex.
- **Liftoff:** $l \ge l_0$
Contact wrench vanishes; rotors primary for attitude.
- **Handoff**
Events switch authority; no hardware or gain change.

Leg authority exists only in stance; rotor attitude authority is continuous through the full hop.

6 Control - leg-priority hierarchical force allocation



500 Hz architecture: commands + feedback → planner → two-stage allocator → leg torques + rotor thrusts.

- **A. Vertical energy hopping (stance)**

$$E_{sys} = \tfrac{1}{2}m\dot{p}_z^2 + mg\,p_z, \quad E_{des} = mg\,h_{des}$$
$$f^{des}_{c,z} = \max\big(0,\ k_z(h_{des} - p_z) - b_z\dot{p}_z + k_E(E_{des} - E_{sys})\big)$$

Injects push-off energy, absorbs impact.

- **B. SO(3) attitude PD (both phases)**

$$\boldsymbol{e}_R = \tfrac{1}{2}\left(\boldsymbol{R}_{des}^\top\boldsymbol{R} - \boldsymbol{R}^\top\boldsymbol{R}_{des}\right)^\vee$$
$$\boldsymbol{\tau}^{des}_b = -\boldsymbol{K}_R\boldsymbol{e}_R + \boldsymbol{K}_\omega(\boldsymbol{\omega}^{des} - \boldsymbol{\omega})$$

Roll/pitch only; yaw left free. $\boldsymbol{\tau}^{des} = \boldsymbol{R}\boldsymbol{\tau}^{des}_b$.

- **C. Leg first - constrained least squares**

$$\boldsymbol{f}^{cmd}_{c,xy} = \arg\min_{\boldsymbol{f}_{xy}} \left\|(\boldsymbol{r}_c\times\boldsymbol{f}_c)_{rp} - (\boldsymbol{\tau}^{des})_{rp}\right\|_2^2$$
$$\text{s.t. } \|\boldsymbol{f}_{xy}\|_2 \le \mu f^{des}_{c,z}, \quad \|\boldsymbol{J}^\top(-\boldsymbol{R}^\top\boldsymbol{f}_c)\|_\infty \le \tau_{max}$$

The leg realizes all it can under friction and torque limits.

- **D. Rotors take only the residual**

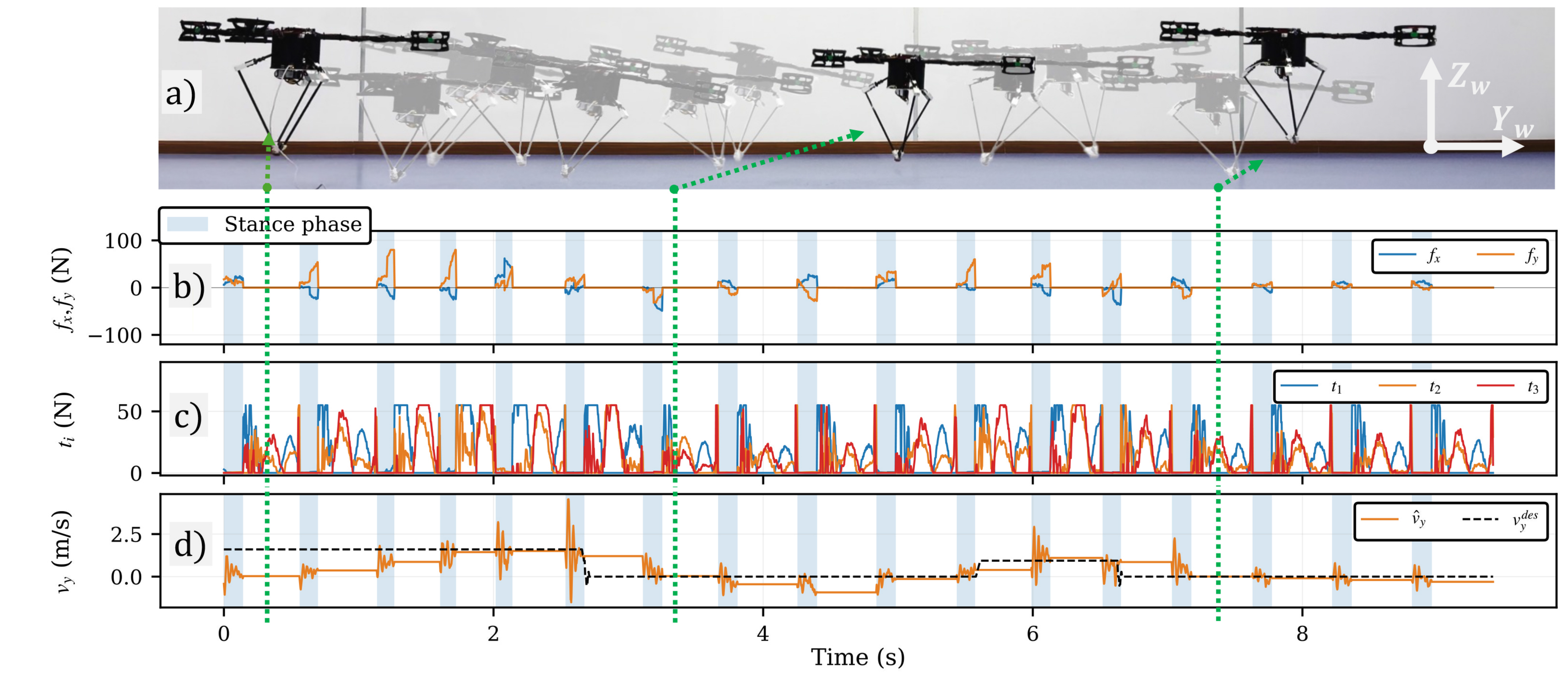
$$\boldsymbol{\tau}^{cmd}_{res} = \boldsymbol{\tau}^{des} - \boldsymbol{r}_c\times\boldsymbol{f}^{cmd}_c$$
$$\delta\boldsymbol{t}^{cmd} = \arg\min_{\delta\boldsymbol{t}} \left\|(\boldsymbol{M}_t\,\delta\boldsymbol{t})_{rp} - (\boldsymbol{\tau}^{cmd}_{res})_{rp}\right\|_2^2$$

s.t. thrust bounds; $\boldsymbol{t}^{cmd} = \frac{T_{base}}{3}\mathbf{1}_3 + \delta\boldsymbol{t}^{cmd}$, $T_{base} = 0.02\,mg$.

- **E. Flight - rotors primary; Raibert-style foot placement**

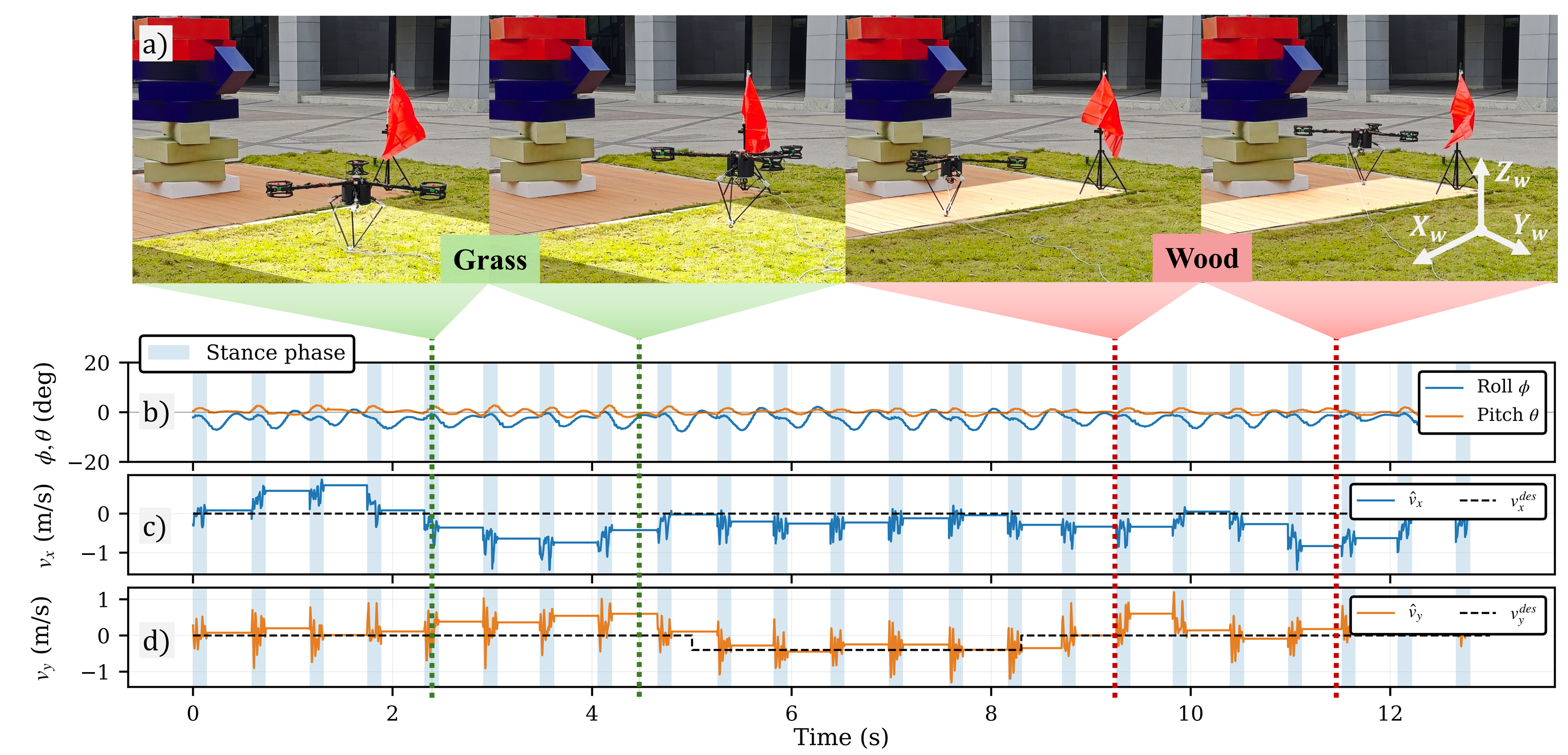
$$\boldsymbol{f}_c = \boldsymbol{0}; \quad \text{rotors track } (\boldsymbol{\tau}^{des})_{rp}; \quad \boldsymbol{r}^{des}_{td,xy} = k_v\boldsymbol{v}_{xy} - k_r\boldsymbol{v}^{des}_{xy}$$
$$\boldsymbol{p}^{des}_{foot} = \left[x^{des}_{td},\ y^{des}_{td},\ -\sqrt{l_0^2 - \|\boldsymbol{r}^{des}_{td,xy}\|_2^2}\right]^\top$$

7 Validation - indoor stop-and-go tracking



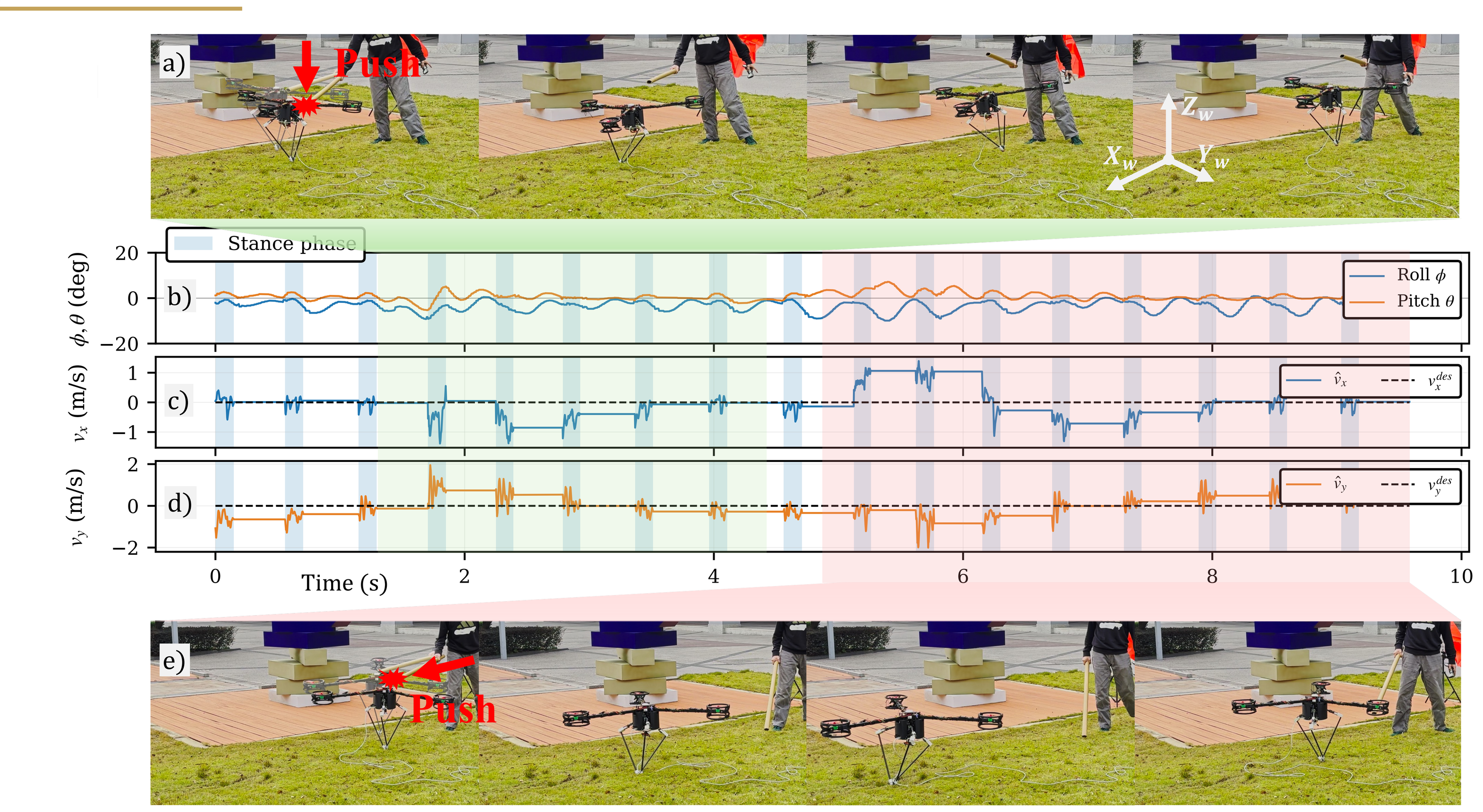
Stop-and-go over 17 hops: velocity recovered stride to stride by Raibert foot placement.

8 Validation - grass to wood



Grass → wood, 23 hops: same controller - no terrain model, no gain retuning.

9 Validation - push recovery



Rotor-arm and body pushes rejected - stable tracking again in 1–3 hops.